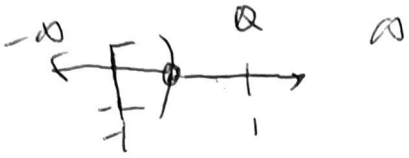
Informal 21: UpperBound but no max

Ideas: $\mathbb{N}$, $\{1,2,3\ldots\}$, $[-1,0) \subset \mathbb{Q}$, $[0,1)$



0 is an upper bound with no maximum

$\mathbb{N}^+$ and $\{1,2,3\ldots\}$ have no upper bound

Ex 19) Proof: Let S be a set with minimum m & m'
We see $m, m' \in S$ by definition. Further, $m \leq x \ \forall \in S$
In particular $m \leq m'$. On the other hand $m' \leq x \ \forall \in S$.
So, $m' \leq m$. By trichotomy they're equal. $m = m'$

Thm 29: Every upper bounded non empty subset of the naturals has a max

Proof Sketch: Let t be such a set, and write B for an upperbound. Thus if $x \in t$, then $x \leq B$. Likewise $x \geq 0$ since $t \subset \mathbb{N}$. We see $T \subset \{0, \ldots, B\} = \{n \in \mathbb{N} : 0 \leq n \leq B\}$. Thm 28 says t has a maximum.

Def. An ordered set U is well ordered if every non-empty subset has a minimum

Thm 30: The natural numbers are well-ordered. "The well ordered principle"

Proof: Let $S \subseteq \mathbb{N}$ be a nonempty subset. Let L be the set of lower bounds of S. NOTE: 0 is in L since $0 \leq x$ in $\mathbb{N}$. In particular, $L \neq \emptyset$
Since $S \neq \emptyset$ there exists some element in S. If $l \in L$, then $l \leq k$ since l is a lower bound. We see k is an upper bound for L.
Thm 29 says L has a max; call it M. We wish to show $M \in S$.
For a contradiction, suppose M is not in S. Thus M is $< k$ for every k in S. By Thm 14, $M+1 \leq k$ for every $k \in S$. $M < k < M+1$. Thus $M+1 \in L$ and $m < m+1$ contradicting the maximality of M. $\cancel{\times}$ $M \in S$